

The empirical recurrence for OEIS A251563: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A251563 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A251563 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock summing to a prime.”. It has offset 1 and begins

29476, 1720558, 100477236, 5871247808, 343048858208, 20045964354544, 1171337430239996, 68446666138149356

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:27:41, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	87	c_{22}	−14694128078486685901760464491209	c_{43}	5
c_2	5637	c_{23}	70248862561608972894377253644703	c_{44}	−5
c_3	−645435	c_{24}	1214287054289817111215438576349027	c_{45}	−4
c_4	−7308331	c_{25}	−7401030701618268267793781218587325	c_{46}	63
c_5	1811307789	c_{26}	−75166737854797990583027083195608980	c_{47}	34
c_6	−8646932882	c_{27}	551456771059437790407748745748595084	c_{48}	−5
c_7	−2666681287794	c_{28}	3508697525423524321253585347423000440	c_{49}	−19
c_8	34305884448653	c_{29}	−30479458719293167919193132296574163952	c_{50}	359
c_9	2322687500939895	c_{30}	−123623890603353804765507136185772538272	c_{51}	924
c_{10}	−43039805868269789	c_{31}	1285394624699001972755136799266453022512	c_{52}	−19
c_{11}	−1257308237539496586	c_{32}	3255260315579791568505327788466096017568	c_{53}	−34
c_{12}	30233024857809226308	c_{33}	−42166204813406476002476007363836386763200	c_{54}	809
c_{13}	426272322083409995445	c_{34}	−61819343422423686861648338817197120559744	c_{55}	1024
c_{14}	−13514743621181899102289	c_{35}	1091146060309787111270620657457248659747168	c_{56}	−274
c_{15}	−85938261878757981103960	c_{36}	748145099635326247218132761527120778238784	c_{57}	−235
c_{16}	4084044939277441618424815	c_{37}	−22507390099342965960391021002728406388996096	c_{58}	7397
c_{17}	7362738578286459424024420	c_{38}	−2097228587266835954690902550648320507140864	c_{59}	4045
c_{18}	−865505814084252734513175996	c_{39}	372942055898493751986772482723645058367462144	c_{60}	−1579
c_{19}	991560520844541700530720456	c_{40}	−138531608375975256371951227574557574213881600	c_{61}	−482
c_{20}	131844618669655969001357550429	c_{41}	−4991310772415490068225396844015282383534309120	c_{62}	2641
c_{21}	−423382666753653975705461032739	c_{42}	3730640971398773573330581981603630811043230720	c_{63}	3091

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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